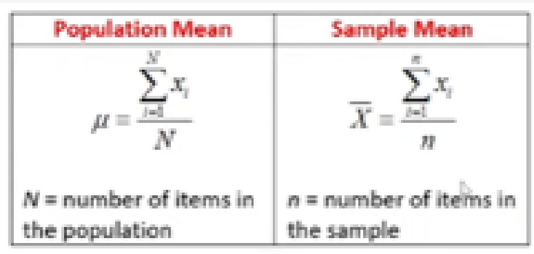
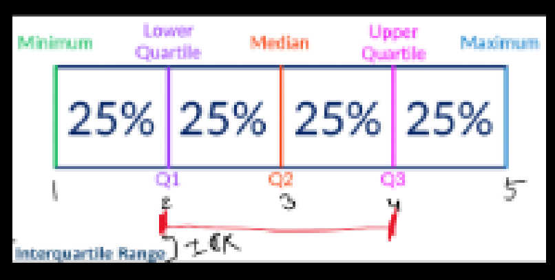
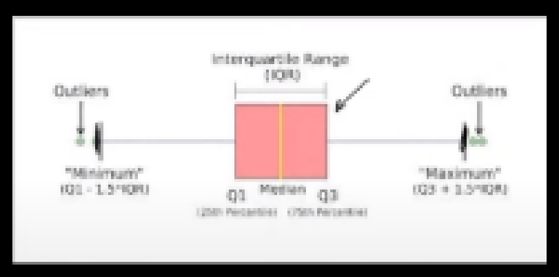
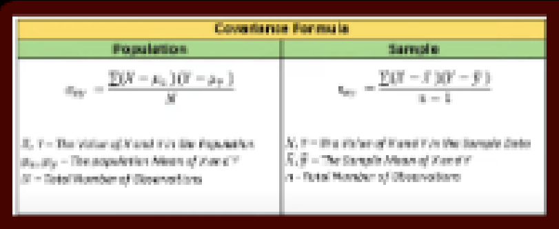
Statistics-Cheat-Sheet

- Types
  - Descriptive
  - Inferential
- population and sample
- Descriptive Statistics
  - Describe the data
  - Measure of central tendency
    - mean
      - effectived by outliers
      - sum all value / total number of rows
      - 

Population Mean	Sample Mean
$\mu = \frac{\sum_{i=1}^{N} x_i}{N}$	$\overline{X} = \frac{\sum_{i=1}^{n} x_i}{n}$
N = number of items in the population	n = number of items in the sample

        Formula
    - median
      - less effictive by outliers
      - sort value , take the middle value
      - 50 percentile
    - mode
      - used for categorical columns, most frequest category
  - Measure of dispersion (how spread out is the data)
    - Range
      - min - max
      - effected by outliers
    - variance
      - $$\sigma^2 = \frac{\Sigma (x-\mu)^2}{N} \quad \text{Population Variance}$$
        $$s^2 = \frac{\Sigma (x-\bar{x})^2}{n-1} \quad \text{Sample Variance}$$
        Average of squared differences from mean.
      - **Used to Tell the spread of the data, but it is squared and not in same unit SD is uded**
    - Mean absolute devaition (MAD)
      - not differentialbe , so not able to help in prediction, not used mostly
      - $$MAD = \frac{\sum |x_i - \overline{x}|}{n}$$
        Same like variance but it does not use square, it use mod.
    - standard diavtion
      - √varience
      - Same unit as the data
    - Cofficient of variation (CV)
      - Formula:
        $$CV = \frac{\text{Standard Deviation (SD)}}{\text{Mean}} \times 100$$
        Compare the dispersion of 2 different columns
    - Percentiles / 5 Number summary
      - To understand 5 point summary we need to know percentile
      - Percentile
        - Divide the data into 100 equal parts
        - if you get 99% percentile that mean 99 percent of people got less amrks then you
        - It is not compulsory that a percentile value is present in the data
      - 5 number summary
        - Minimum value
        - First quartile (Q1) - 25 Percentile
        - Median (Q2) - 50 Percentile
        - Third quartile (Q3) - 75 Percentile
        - Maximum value
        - **Between 25 to 75 percentile is called IQR (Interquartile range)**
        - Note
          - 

            The five-number summary is often represented visually using a **box plot**
          - 

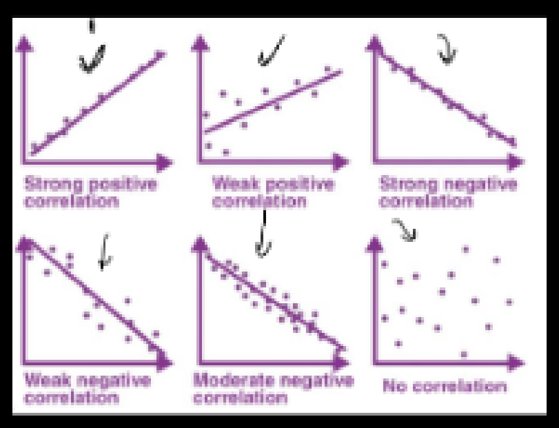
            BOX PLOT
  - Covarience and correlation
    - Covarience
      - 

Covariance Formula	
Population	Sample
$\sigma_{XY} = \frac{\sum(X-\mu_X)(Y-\mu_Y)}{N}$	$s_{XY} = \frac{\sum(X-\overline{X})(Y-\overline{Y})}{n-1}$

        It tells us how two variables change together.
      - **If covariance value is positive = positive relationship,**

        If covariance value is negative = negative relationship

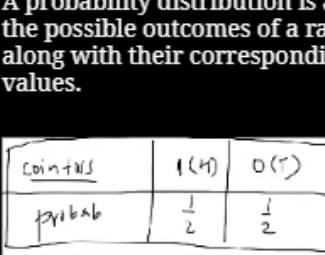
        If covariance value is zero = no relationship
      - **One limitation of covariance is that it does not tell us about the strength of the relationship between two variables, since the magnitude of covariance is affected by the scale of the variables.**
    - correlation
      - $$Correlation = \frac{Cov\,(x,y)}{\sigma x * \sigma y}$$
        Correlation tells us how strong and in what direction two variables are related.

        It is like covariance but with magnitude
      - 

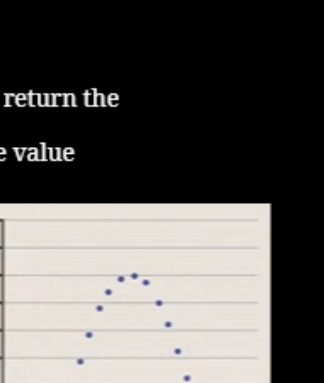
        Its value always lie in between -1 t0 1
        - if value is negative = that mean negative correlation
        - if value is positive = that mean positive correlation
        - if value is 0 = that mean no correlation
        - more the value closer to 1 -> mean stronger + correlation
        - more the value closer to -1 -> mean stronger - correlation

Presented with xmind

Statistics-Cheat-Sheet

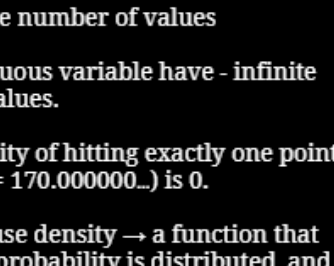
- **Random variable**
  - A Random Variable is a set of possible values from a random experiment.
  - Types
    - Discrete Random Variable — Example - dice tose experiment - Random Variable = {1,2,3,4,5,6}
    - Continuous Random Variable — Example - CGPA of student experiment - Random Variable = { 0 , ...2.3,.... 10}
- **Probability Distributions**
  - A probability distribution is a list of all of the possible outcomes of a random variable along with their corresponding probability values.
  - 

    Coin toss
    P(H)
    P(T)
    probab
    1/2
    1/2

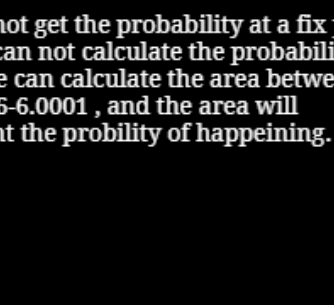
    Example - probability Distribution for a coin toss
  - **But for a complex example, we can create the table by hand so, we have decided to make/find a function which will return the probability for a value of a random variable and that function is Probability Distribution Function**
- **Probability Distribution Function**
  - Function that take a value of a random variable and return the
    1 - for descrete case(PMF) - probability of that value
    2- For continuous case(PDF) - probability density at the value
  - Types
    - Probability Mass function (PMF)
      - 
        A function that take value of a discrete random variable and return probability of that value
      - Example
        - Bernoulli distribution
        - Binomial distribution
    - Probability density function (PDF)
      - A function that take value of a continuous random variable and return probability density at that value
      - Example
        - normal distribution
        - log normal distribution
        - poisson distribution
      - area under the curver = 1
      - **Q1 -Why Probability Density and why not Probability? on y - axis**
        - Unlike the discrete random variable which has countable number of values

          In the continuous variable have - infinite number of values.

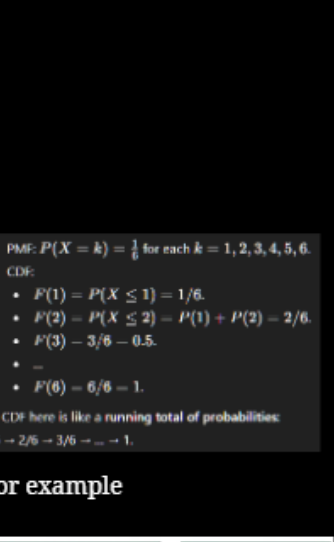
          The probability of hitting exactly one point (e.g., height = 170.000000...) is 0.

          Instead, we use density → a function that tells us how probability is distributed, and we get real probabilities by taking areas under the curve (integrals).
      - **Q2 -What does the area of this graph represents?**
        - Probability!!
          - 

            Example- probability of getting marks in between 5-6 = will be tell by calculating the area under probability density(5-6). that will be the probability
        - we can not get the probability at a fix point like we can not calculate the probability at 6, but we can calculate the area between 6-6.01 or 6-6.0001 , and the area will represent the probility of happeining.
      - **Note -> if you integrate Probability density function (PDF) it will give you Cumulative Distribution Function(CDF)**

        and if you differentiate Cumulative Distribution Function(CDF) it will give you Probability density function (PDF)
      - The Total area under the curve = 1
    - Cumulative distribution function (CDM)
      - For PMF - this function take a value of discrete random variable and return the sum of probability up to the point/value
        - 
          - PMF: $P(X=k) = \frac{1}{6}$ for each $k = 1,2,3,4,5,6$.
          - CDF:
            - $F(1) = P(X \le 1) = 1/6$.
            - $F(2) = P(X \le 2) = P(1) + P(2) = 2/6$.
            - $F(3) = 3/6 = 0.5$.
            - ...
            - $F(6) = 6/6 = 1$.

          So, CDF here is like a running total of probabilities:
          1/6 → 2/6 → 3/6 → ... → 1.

          For example
      - Cumulative Distribution Function(CDF) of PDF
        - 

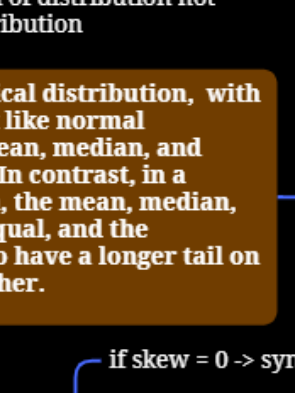
          Example - rolling a dice ->

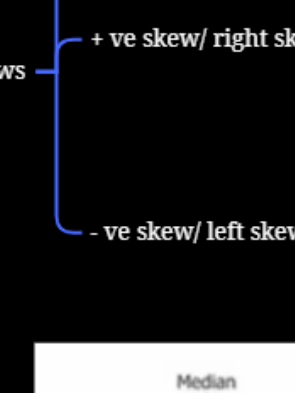
          F(4) = p(4)+p(3)+p(2)+p(1)
- **Density Estimation**
  - why we are learning Density Estimation — To figure out the shape of the probability distribution (PDF) from data.-
  - types
    - Parametric DE
      - Here we assume the data follows a specific distribution shape (like Normal, Exponential, etc.).

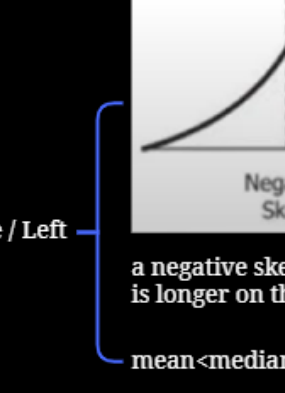
        Then we only need to estimate the parameters (like mean μ and variance σ² for Normal distribution).

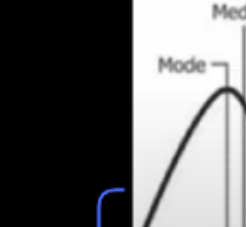
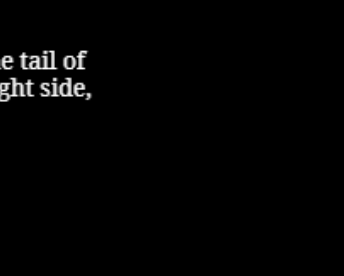
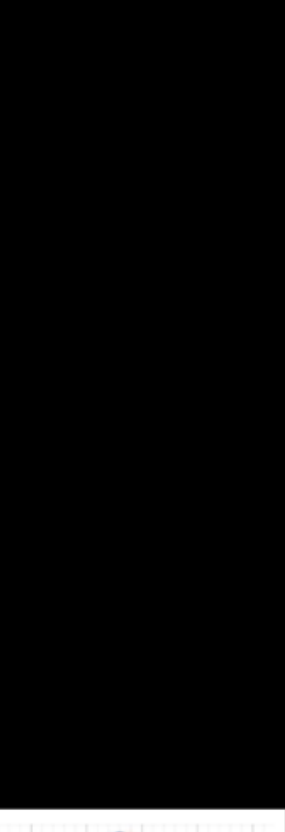
        Once parameters are found, the whole distribution (PDF) is defined.
    - Non - Parametric DE
      - Here we don't assume the data follows a specific distribution shape (like Normal or Exponential).

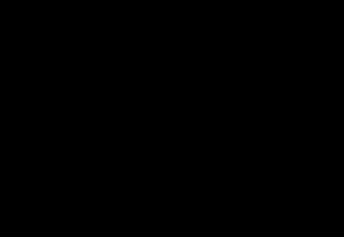
        Instead, we let the data itself decide the shape.

        It's more flexible but can be more complex.
  - Common techniques
    - **Kernel density estimation (KDE)** — **skipped**
    - histogram estimation
    - Gaussian mixture models (GMMs)
- **Skewness**
  - Skewness is a measure of the asymmetry of a probability distribution.
  - It can be for any kind of distribution not only for Normal distribution
  - **If a pdf is symmetrical distribution, with unimodel (one peek like normal distribution) the mean, median, and mode are all equal. In contrast, in a skewed distribution, the mean, median, and mode are not equal, and the distribution tends to have a longer tail on one side than the other.** — **The greater the skew the greater the distance between mode, median and mean.**
  - Different values of skews
    - if skew = 0 -> symetrical values
    - + ve skew/ right skew
      - If skew = 0.5 — a littel bit of +ve skew / right skew
      - If skew = 1 — more +ve skew / right skew
      - If skew = 1.5 — more +ve skew / right skew
    - - ve skew/ left skew
      - If skew = -0.5 — a littel bit of -ve skew / left skew
      - If skew =- 1 — more -ve skew / left skew
      - If skew = -1.5 — more -ve skew / left skew
  - Types
    - -ve / Left
      - 

        a negative skewness means that the tail is longer on the left side.
      - mean<median<mode
    - +ve / Right
      - 

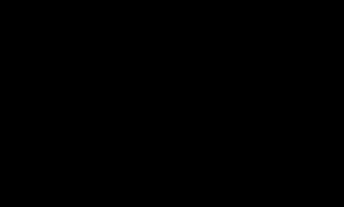
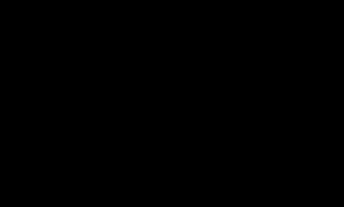
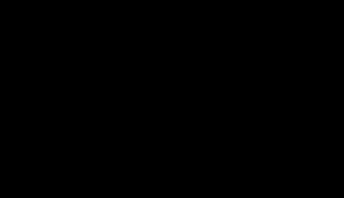
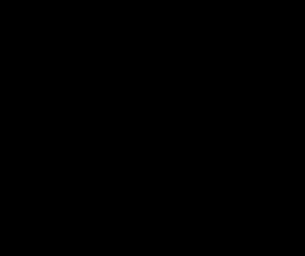
        A positive skewness means that the tail of the distribution is longer on the right side,
      - mode<median<mean
    - 0
      - 

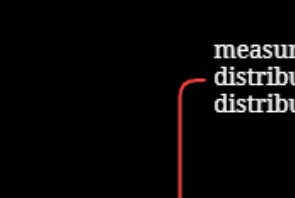
        A zero skewness indicates a perfectly symmetrical distribution.
- **Normal Distribution / Gaussian distribution**
  - 
    It is a Probability density function (PDF) -> And for a function we can draw a graph.
  - very common in nature
  - **Properties of Normal Distribution**
    - 
      1. Symmetricity
    - 

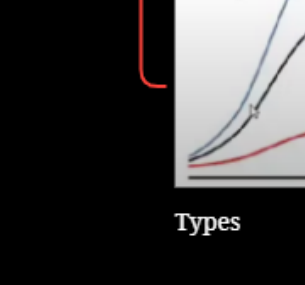
      mean == median == mode , For a proper Normal distribution
    - 
      3. Empirical Rule
      - About 68.27% of values lie within 1 standard deviation of the mean
      - About 95.45% within 2 standard deviations
      - About 99.73% within 3 standard deviations
  - $$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$
    PDF Function

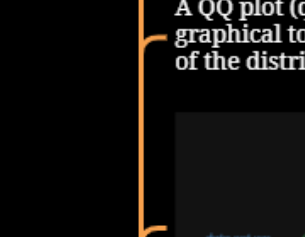
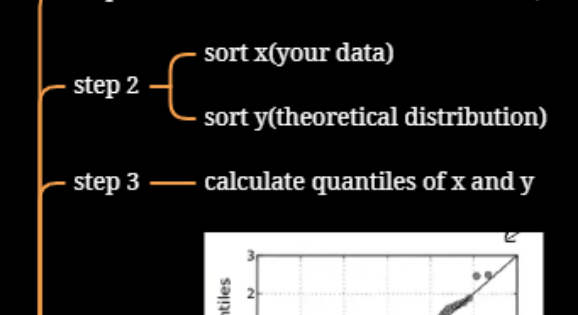
    x -> value for what we want to get the probability density,

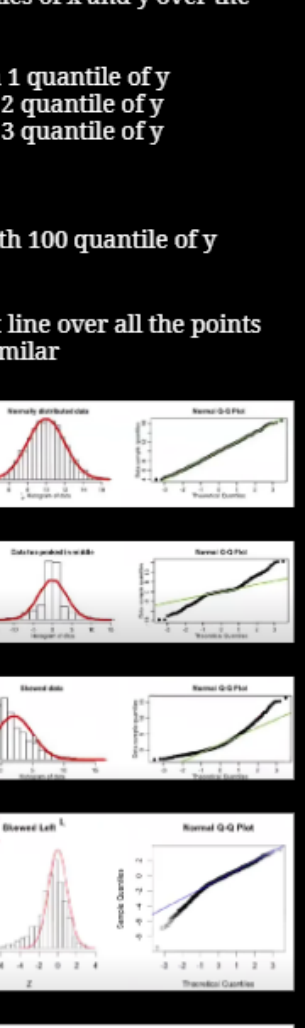
    σ -> is standard deviation

    μ-> is mean
    - how changing the mean effect the Graph
      - 
        Increasing the mean -> will lead to movement of the complete graph toward the positive direction of the x-axis
      - 
        Decreasing the mean -> will lead to movement of the complete graph wtoward the negative direction of the x-axis
      - Changes in mean does not effect the shape of the grph, only the position of the graph
    - how changing the std effect the Graph
      - 
        Increasing the std value make the bell curve more disperse and decreasing the std value will lead to make values come closer to the mean
      - Changes in std effect the shape of the graph.
  - 
    CDF curve
    - $$F(x) = P(X \le x) = \int_{-\infty}^{x} f(t)\,dt = \frac{1}{\sigma\sqrt{2\pi}}\int_{-\infty}^{x} e^{-\frac{(t-\mu)^2}{2\sigma^2}}\,dt.$$
      CDF function -> it is just integration of pdf function
  - Standard normal variate (Z) / Standard normal distribution
    - If the σ = 1, μ=0 then this normal distribution is called "Standard normal prior variate"
    - $$\phi(z) = \frac{1}{\sqrt{2\pi}}e^{-z^2/2},$$
      PDF Function
    - $$z_i = \frac{x_i - \mu}{\sigma},$$
      we can convert any normal distribution in standard normal variate (z), using this formula
    - Benefits of standard normal variables — we can compare 2 different distribution
- **Kurtosis**
  - Kurtosis is the 4th statistical moment. — Example -

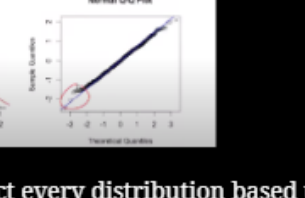
    1 moment = mean
    2 moment = std
    3 moment = skewness
    4 moment = Kurtosis
  - is a measure of the "tailedness" of the probability distribution
  - Example ->
    - 
      here mean is same, skewness is same , standard deviation is also the same but the difference is in between the fatness of the tail of 2 distribution.

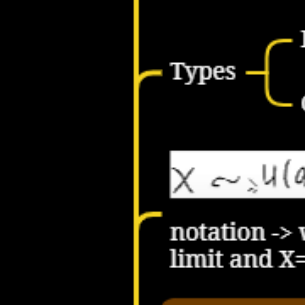
      Red distribution tail is mor fater then black distribution
  - Excess Kurtosis
    - measure of how much more peaked or flat a distribution is compared to a normal distribution,
    - How it is calculate excess kurtosis
      - step1 -
        - considered normal distribution kurtosis as 3.
        - note if the function we are using has kurtosis value of normal distribution as 0 already then we do not need to do the second step.
      - step 2 - — calculate sample kurtosis of the distribution
      - step 3 - — sample kurtosis -3
        - < 0 — Leptokurtic
        - > 0 — Platykurtic
        - = 0 — Mesokurtic
    - 

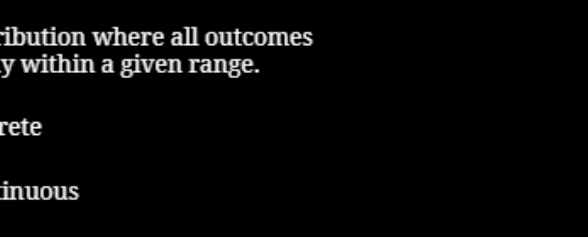
      Types
      - Leptokurtic
        - A distribution with positive excess kurtosis is called leptokurtic.
        - In terms of shape, a leptokurtic distribution has fatter tails.
        - This indicates that there are more extreme values or outliers in the distribution.
      - Platykurtic
        - A distribution with negative excess kurtosis is called platykurtic.
        - a platykurtic distribution has thinner tails.
        - This indicates that there are fewer extreme values or outliers in the distribution.
      - Mesokurtic
        - Distributions with zero excess kurtosis are called mesokurtic.
        - indicating that it has the same degree of "peakedness" or "flatness" as a normal distribution.
- **QQ Plot (quantile-quantile plot)**
  - Used to find a given column follow a normal distribution (or another distribution) or not
  - A QQ plot (quantile-quantile plot) is a graphical tool used to assess the similarity of the distribution of two sets of data.
  - 
    Example
  - Steps
    - step 1 — Get a theoretical distribution(normal)
    - step 2
      - sort x(your data)
      - sort y(theoretical distribution)
    - step 3 — calculate quantiles of x and y
    - step 4
      - 
        start plotting quantiles of x and y over the graph like ->

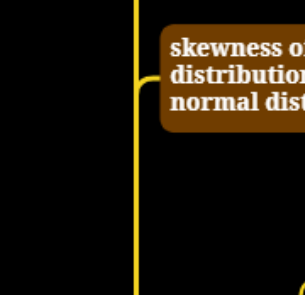
        1 quantile of x with 1 quantile of y
        2 quantile of x with 2 quantile of y
        3 quantile of x with 3 quantile of y
        .
        .
        .
        100 quantile of x with 100 quantile of y
    - step 5 — if there is a straight line over all the points at mean they are similar
  - Interpret QQ plot (normal theoretical distribution)
    - 

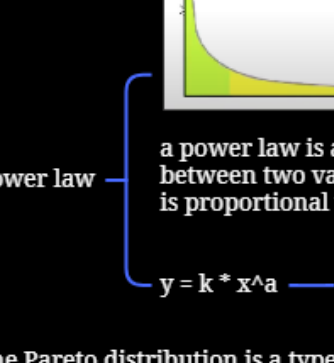
  - Does QQ plot only detect normal distribution? — No, it detect every distribution based upon the theoretical distribution you have choose
- **Uniform distribution**
  - probability distribution where all outcomes are equally likely within a given range.
  - Types
    - Discrete
    - Continuous
  - $X \sim U(a,b)$

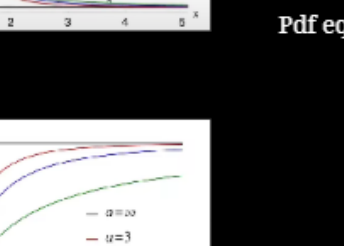
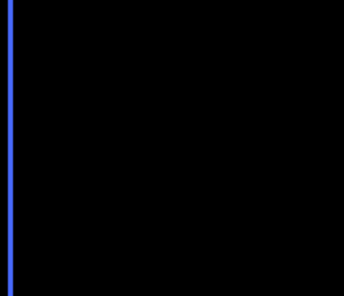
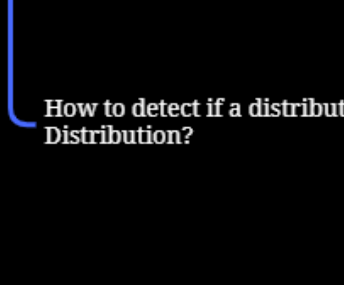
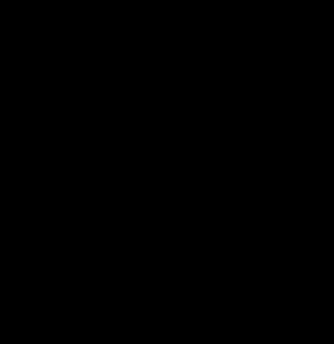
    notation -> where a,b = upper and lower limit and X= data set
  - **skewness of continuous uniform distribution = 0 = it is symmetrical like normal distribution**
  - Example
    - discrete uniform distribution
      - 

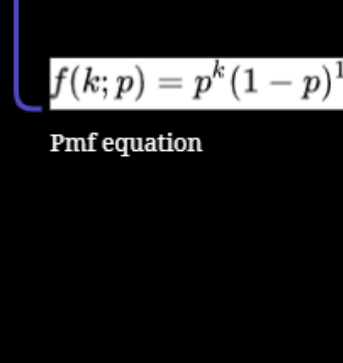
        throwing a dice = probability of each outcome = 1/6
    - continuous normal distribution — The height of a person randomly selected from a group of individuals whose heights range from 5'6" to 6'0"
  - 

    pdf
    - 

      The area of pdf =1
      Hight * base = 1
      Hight = 1/base
      Hight = 1/(b-a)
    - $f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \le x \le b \\ 0 & \text{for otherwise} \end{cases}$
  - 

    cdf
  - **Example**
    - Random initialization: — In many machine learning algorithms, such as neural networks and k-means clustering, the initial values of the parameters can have a significant impact on the final result. Uniform distribution is often used to randomly initialize the parameters, as it ensures that all values in the range have an equal probability of being selected.
    - Sampling: — Uniform distribution can also be used for sampling. For example, if you have a dataset with an equal number of samples from each class, you can use uniform distribution to randomly select a subset of the data that is representative of all the classes.
    - Hyperparameter tuning: — Uniform distribution can also be used in hyperparameter tuning, where you need to search for the best combination of hyperparameters for a machine learning model. By defining a uniform prior distribution for each hyperparameter, you can sample from the distribution to explore the hyperparameter space.
- **Log normal distribution**
  - 

    lognormal distribution is a heavy tailed continuous probability distribution of a random variable whose logarithm is normally distributed.
  - **a right skewed distribution is called log normal distribution , if we take log of the data makes a normal distribution**
  - Skewed distribution
  - How to detect if a distribution is log normal Distribution? — if the log value of a column is normally distributed [how to check normal distribution = QQ plot]
- **Pareto distribution**
  - Power law
    - 
      a power law is a functional relationship between two variables, where one variable is proportional to a power of the other.
    - y = k * x^a — where k= some random constant y,x= variable and a = power
  - The Pareto distribution is a type of probability distribution that is commonly used to model the distribution of wealth, income, and other quantities that exhibit a similar power-law behaviour
  - 
    pdf
    - $$\frac{\alpha x_m^\alpha}{x^{\alpha+1}}$$
      Pdf equation
  - 
    cdf
  - Skewed distribution
  - How to detect if a distribution is Pareto Distribution?
    - by the help of log log plot
      - y = $\frac{\alpha x_m^\alpha}{x^{\alpha+1}}$
      - 

        take log of both x and y and then plot a graph of it , if looks like this then it is a pareto distribution
    - QQ plot — instant of taking a normal distribution take pareto distribution and then compare both of them
- **Bernoulli Distribution**
  - Bernoulli distribution is a discrete probability distribution that models a binary outcome, where the outcome can be either success (represented by the value 1) or failure (represented by the value 0).
  - Example - coin toss = either head or tail
  - 

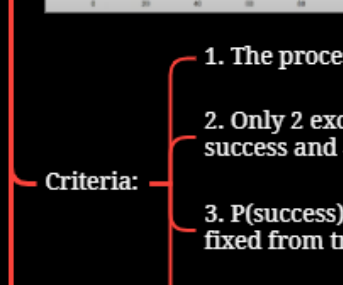
    Pmf graph
  - $f(k;p) = p^k(1-p)^{1-k}$

    Pmf equation
    - this distribution has only one parameter which p= probability of success for the case(k)
    - here k= out come you want , p = probability of k
    - Example -
      - $P(X=1) = \left(\frac{1}{2}\right)^1\left(\frac{1}{2}\right)^0$

        coming 1 on toss
      - $P(X=0) = \left(\frac{1}{2}\right)^0\left(1-\frac{1}{2}\right)^{1-0} = 1 \times \frac{1}{2}$

        coming 0 on toss
- **Binomial Distribution**
  - The binomial distribution is characterized by two parameters: the number of trials n and the probability of success p.

    Binomial distribution is a discrete probability distribution that describes the number of successes in a fixed number of independent Bernoulli trials with two possible outcomes (often called "success" and "failure"), where the probability of success is constant for each trial.
  - Example
    - **deciding a mail is spam is Bernoulli Distribution but it deciding a mail is spam on the basis of 10 mails is Binomial Distribution**
    - **if the number of trials n — 1 and then this Binomial Distribution will become bernoulli distribution**
  - Formula
    - $P(X) = {}_nC_x\, p^x(1-p)^{n-x}$

      n = number of tiles
      p = probability of success
      x = desired result
  - The Probability of anyone watching this lecture in the future and then liking it is 0.5. What is the probability that:
    - No-one out of 3 people will like it
    - 1 out of 3 people will like it
    - 2 out of 3 people will like it
      - 3C2*(0.5)^2*(0.5)^(3-2)
      - 3/8
    - 3 out of 3 people will like it
  - 
  - Criteria:
    - 1. The process consists of n trials
    - 2. Only 2 exclusive outcomes are possible, a success and a failure.
    - 3. P(success) = p and P(failure) = 1-p and it is fixed from trial to trial
    - 4. The trials are independent.
  - usage
    - Binary classification problems: — In binary classification problems, we often model the probability of an event happening as a binomial distribution. For example, in a spam detection system, we may model the probability of an email being spam or not spam using a binomial distribution.
    - Hypothesis testing: — In statistical hypothesis testing, we use the binomial distribution to calculate the probability of observing a certain number of successes in a given number of trials, assuming a null hypothesis is true. This can be used to make decisions about whether a certain hypothesis is supported by the data or not.
    - Logistic regression: — Logistic regression is a popular machine learning algorithm used for classification problems. It models the probability of an event happening as a logistic function of the input variables. Since the logistic function can be viewed as a transformation of a linear combination of inputs, the output of logistic regression can be thought of as a binomial distribution.
    - A/B testing: — A/B testing is a common technique used to compare two different versions of a product, web page, or marketing campaign. In A/B testing, we randomly assign individuals to one of two groups and compare the outcomes of interest between the groups. Since the outcomes are often binary (e.g., click- through rate or conversion rate), the binomial distribution can be used to model the distribution of outcomes and test for differences between the groups.